Mental Health Days and Socioeconomic Predictors: A Zero-Inflated Negative Binomial Analysis of BRFSS Data

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1. The Data

This analysis uses data from the Behavioral Risk Factor Surveillance System (BRFSS), an annual telephone health survey conducted by the Centers for Disease Control and Prevention (CDC). The BRFSS has been administered since 1984 and is the largest continuously conducted health survey system in the world. Each year, trained interviewers contact adults in all 50 U.S. states, Washington D.C., and U.S. territories to collect data on health behaviors, chronic conditions, and access to preventive care.

The specific dataset used here is a cleaned subset of the BRFSS available on Kaggle, which contains responses from over 400,000 adults and focuses on variables related to mental and physical health outcomes.

> Centers for Disease Control and Prevention. (2011). *Behavioral Risk Factor Surveillance System Survey Data.* U.S. Department of Health and Human Services. Data source: https://www.kaggle.com/datasets/cdc/behavioral-risk-factor-surveillance-system

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2. The Variables

The table below lists the variables used in this analysis. The response variable is the number of days in the past 30 days that a respondent reported poor mental health.

Variable	What It Measures	Units / Values	Type
MENTHLTH	Days of poor mental health in past 30 days	0–30 days	Numeric (response)
PHYSHLTH	Days of poor physical health in past 30 days	0–30 days	Numeric (predictor)

Variable	What It Measures	Units / Values	Type
BMI5CAT	Body mass index category	Underweight / Normal / Overweight / Obese	Categorical (predictor)
INCOME2	Annual household income	8 ordered categories (<$10k to >$75k)	Categorical (predictor)
SLEPTIME	Average hours of sleep per night	Hours	Numeric (predictor)
EXERANY2	Physical activity in past 30 days	Yes / No	Categorical (predictor)
SEX	Sex of respondent	Male / Female	Categorical (predictor)
X_STATE	State of residence	50 U.S. states + D.C.	Categorical (random effect candidate)

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3. The Question

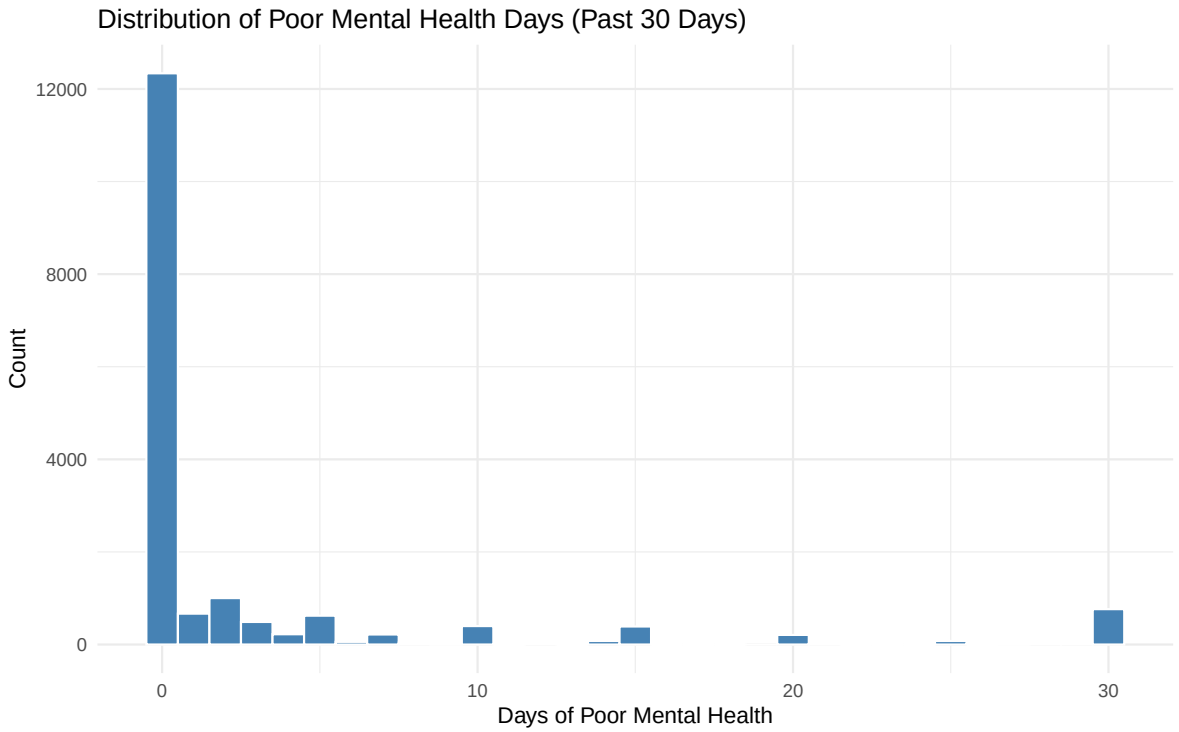
I want to investigate what individual-level health behaviors and socioeconomic characteristics are most strongly associated with poor mental health days among U.S. adults. Specifically, I am interested in whether sleep duration, physical activity, income, and physical health outcomes predict the number of self-reported poor mental health days in a given month, after controlling for sex.

I expected to find that lower income, fewer hours of sleep, physical inactivity, and worse physical health would each be independently associated with more poor mental health days. I also expected geographic variation to be meaningful, though due to computational constraints a mixed-effects model with state as a random effect was not fitted, this is discussed further in the limitations.

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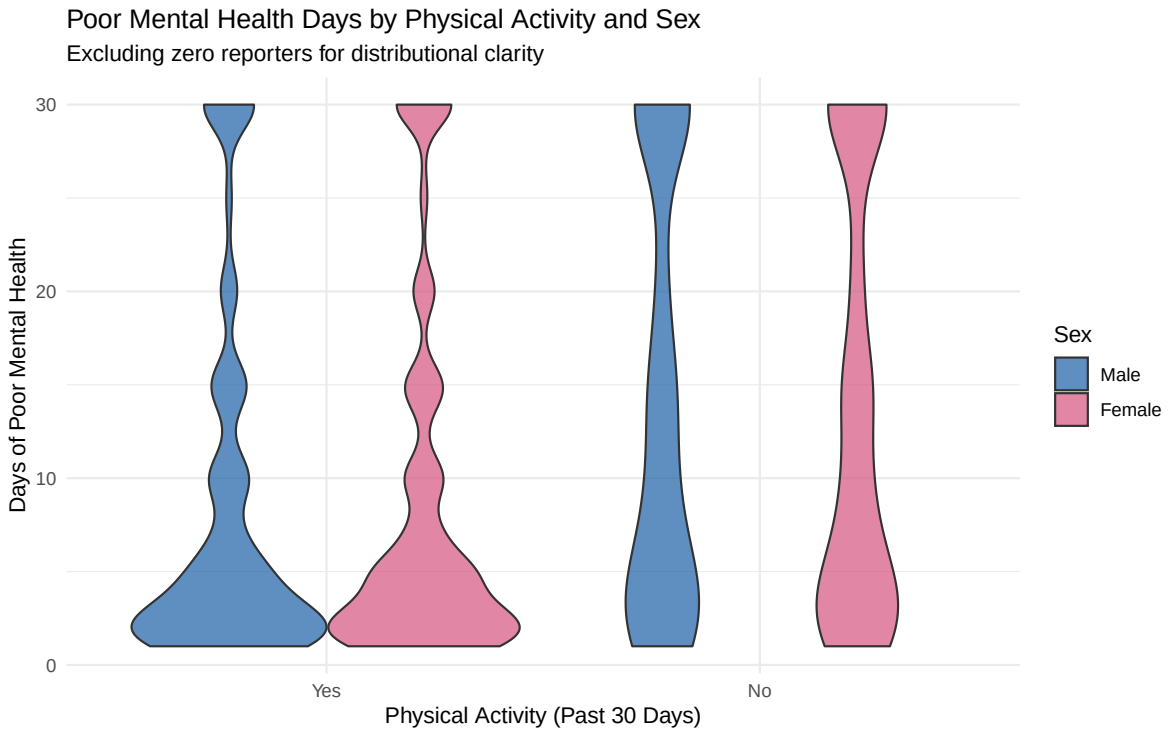
4. Exploratory Analysis

4.1 Distribution of the Response Variable



The response variable is strongly right-skewed with a large concentration of responses at zero, 69.8% of respondents reported zero poor mental health days, and a secondary cluster at 30. This pattern is consistent with zero-inflated count data and rules out ordinary linear regression as an appropriate modeling approach.

4.2 Poor Mental Health Days by Physical Activity and Sex



Respondents who reported physical activity in the past 30 days tended to have fewer poor mental health days than those who did not, and this pattern held for both sexes. Females reported higher poor mental health days on average than males across both activity groups, with inactive females showing the widest spread. This suggested both physical activity and sex would be meaningful predictors, and a potential interaction between the two was worth exploring.

4.3 Overdispersion

Before modeling, I confirmed that the response variable was overdispersed (a necessary condition for preferring negative binomial over Poisson). The mean of MENTHLTH was 2.97 days, while the variance was 50.95, roughly **17× the mean**. This level of overdispersion rules out Poisson regression and motivates the negative binomial family.

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5. Model Selection

5.1 Candidate Models

I fit three candidate models in increasing complexity:

1. **Poisson GLM** — baseline, expected to fail the overdispersion test
2. **Negative Binomial GLM** (`MASS::glm.nb`) — accounts for overdispersion
3. **Zero-Inflated Negative Binomial** (`pscl::zeroinfl`) — accounts for overdispersion and the excess of zeros

All models used the same predictor set:

$$\text{MENTHLTH} \sim \text{INCOME2} + \text{BMI5CAT} + \text{EXERANY2} + \text{SEX} + \text{PHYSHLTH} + \text{SLEPTIME}$$

5.2 Poisson Rejected by Dispersion Test

A formal dispersion test on the Poisson fit (Cam 1990, via `AER::dispersiontest`) returned a dispersion statistic of **33** ($p < 2.2\text{e-}16$), confirming severe overdispersion. The Poisson model was rejected.

5.3 AIC Comparison

Table 2: AIC comparison across candidate models

Model	AIC
Zero-Inflated NB	54813.7
Negative Binomial GLM	55212.2
Poisson GLM	163521.5

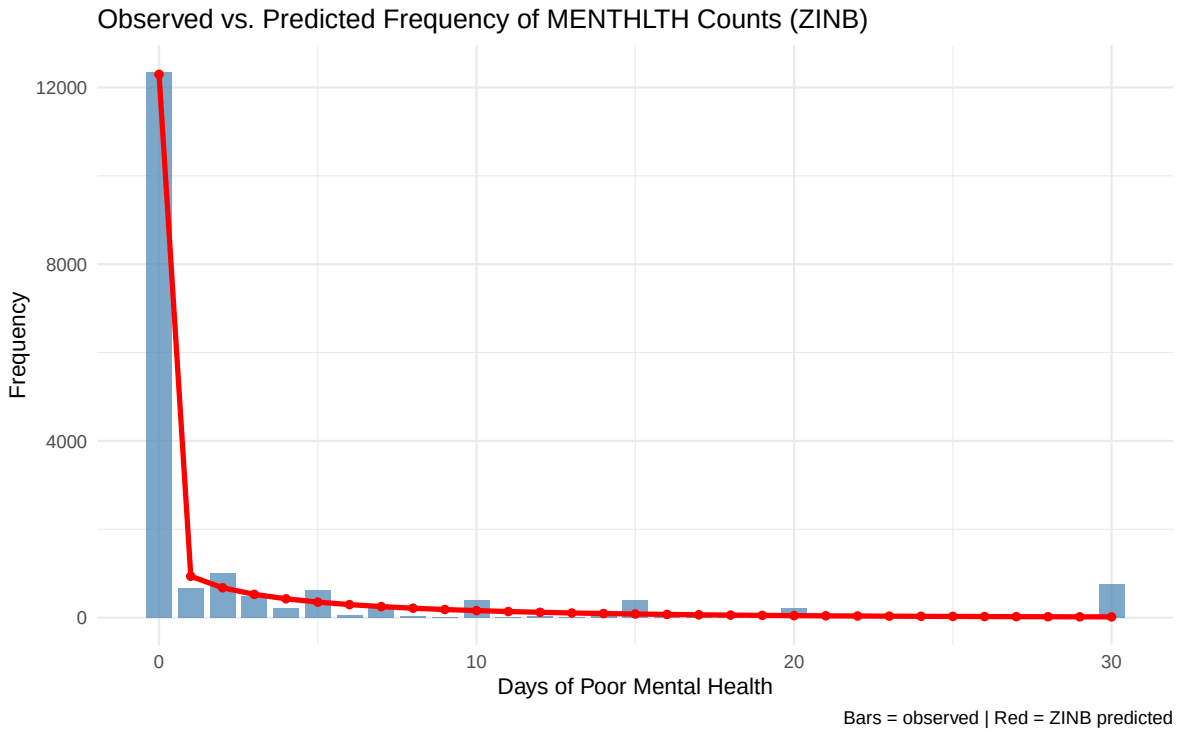
The zero-inflated negative binomial outperformed the plain negative binomial by approximately **400 AIC points**, a decisive margin well above the conventional threshold of 10. This is consistent with the 69.8% zero response rate, which suggests two distinct processes are at play: one governing whether a respondent has *any* poor mental health days, and another governing *how many* they have.

The zero-inflated negative binomial was selected as the final model.

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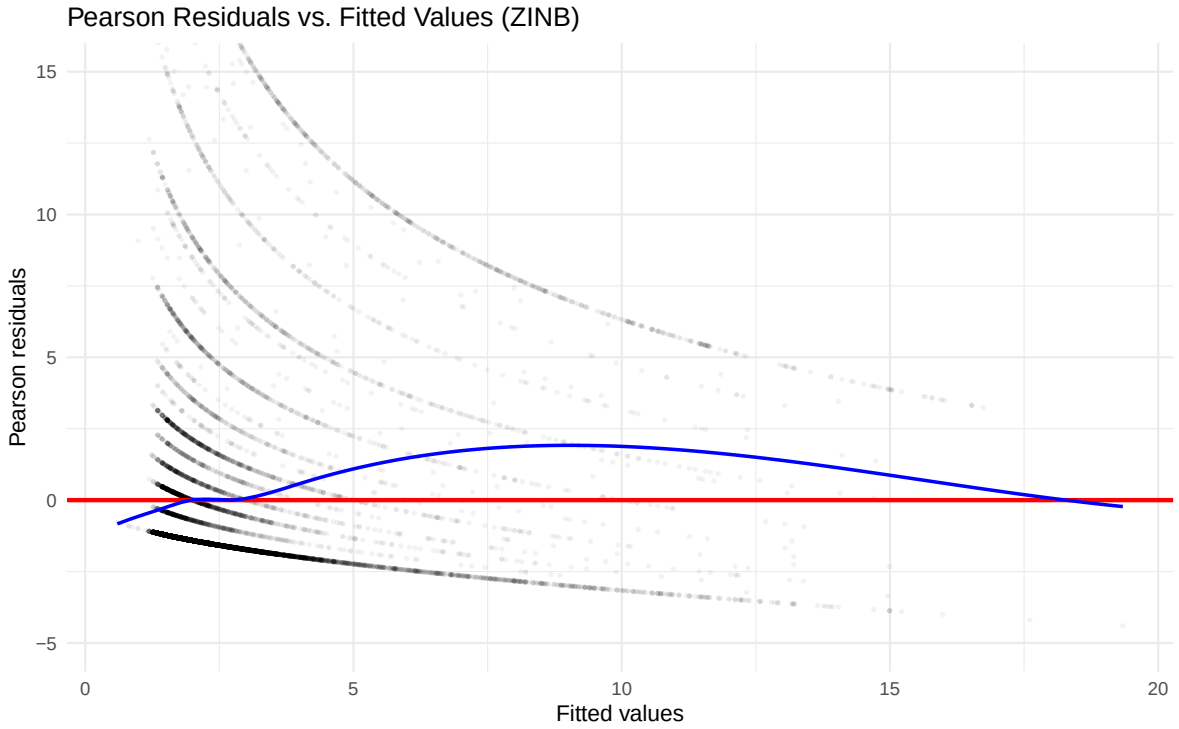
6. Model Diagnostics

6.1 Observed vs. Predicted Counts



The model tracks the observed distribution closely across the full 0–30 range. The zero count in particular is well-captured: observed zeros $= 1.2338 \times 10^4$, predicted zeros $= 1.2297 \times 10^4$. The main limitation is a slight underprediction of the spike at 30 days, likely a ceiling effect from respondents selecting the maximum possible value.

6.2 Pearson Residuals vs. Fitted Values



The curved banding pattern in the residual plot is a known artifact of discrete count data. It reflects the fact that MENTHLTH can only take integer values, and is not indicative of a model failure. The LOESS smoother remains close to zero across the range of fitted values, indicating no systematic bias. The slight hump in the middle indicates that we may slightly be over estimating the people in the middle range.

6.3 Pearson Goodness-of-Fit

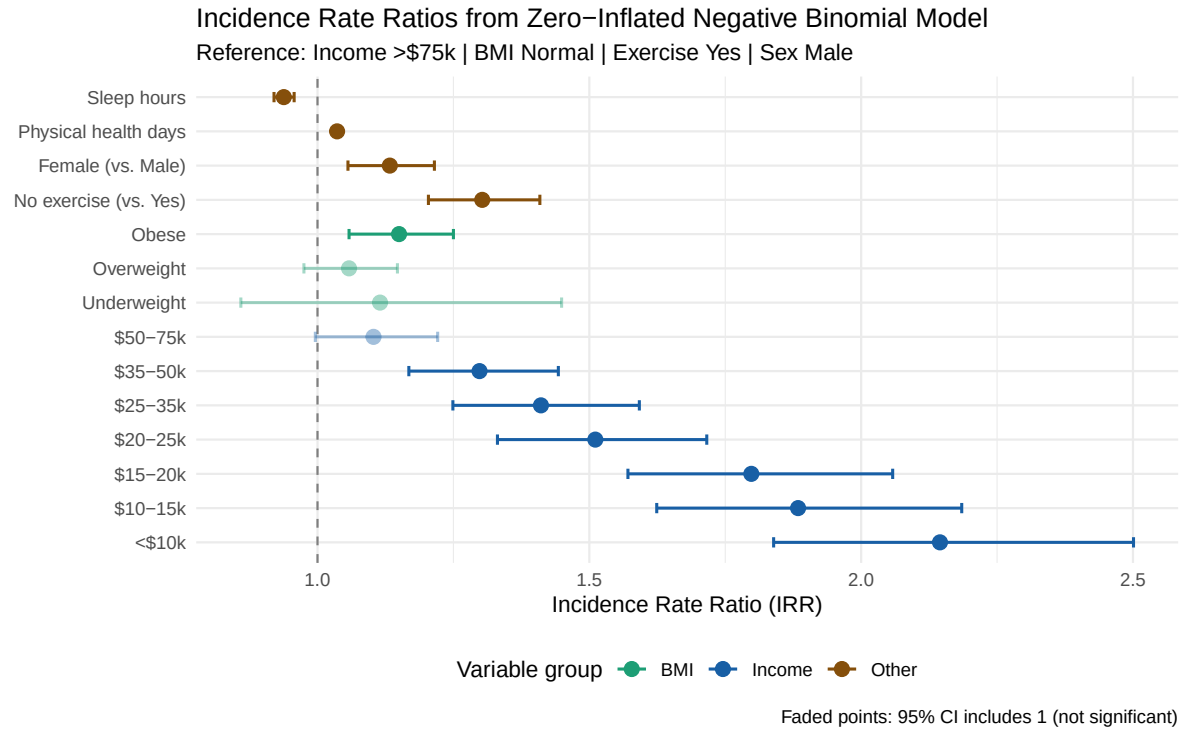
The Pearson goodness-of-fit ratio was **14.68** ($^2 = 2.59372 \times 10^5$, df = 17670). While this is elevated above the ideal of 1.0, it is well-known that with very large samples (n = 17686) even small systematic deviations produce large chi-squared statistics. This does not invalidate inference from the model but is worth noting as a limitation.

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7. Results

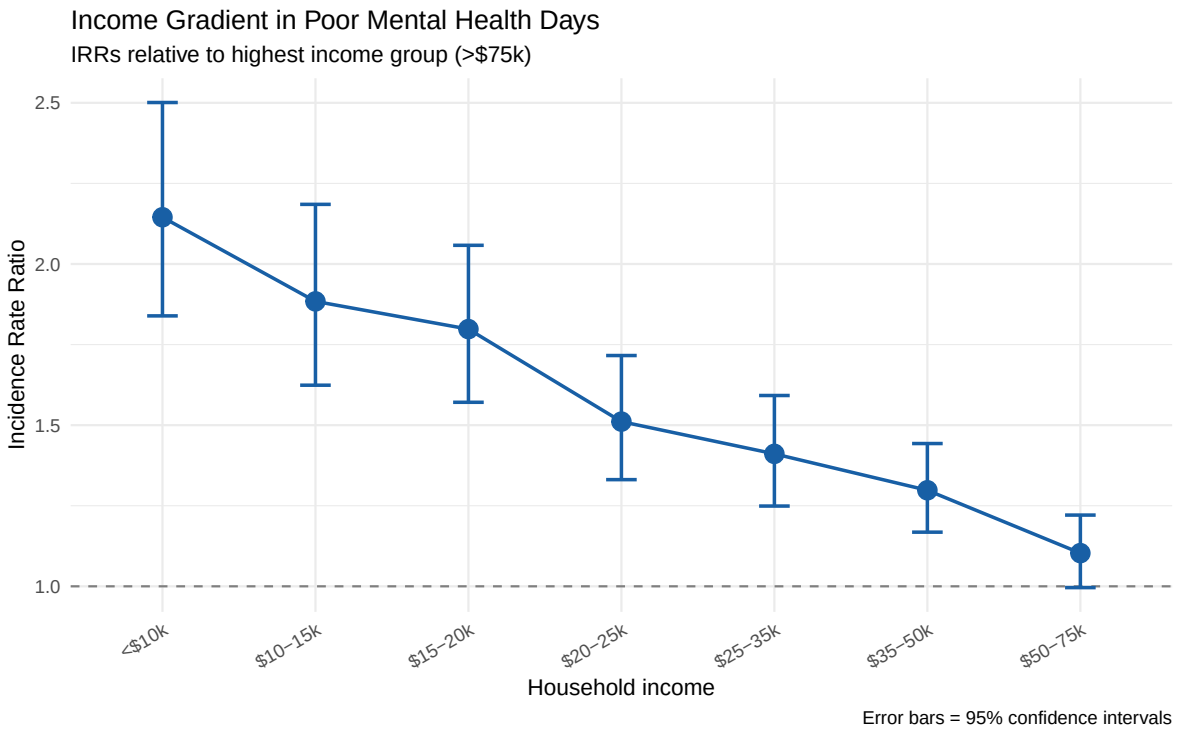
7.1 Incidence Rate Ratios

Coefficients from the count component of the ZINB model were exponentiated to produce incidence rate ratios (IRRs). An IRR greater than 1 indicates more poor mental health days relative to the reference group; an IRR less than 1 indicates fewer.



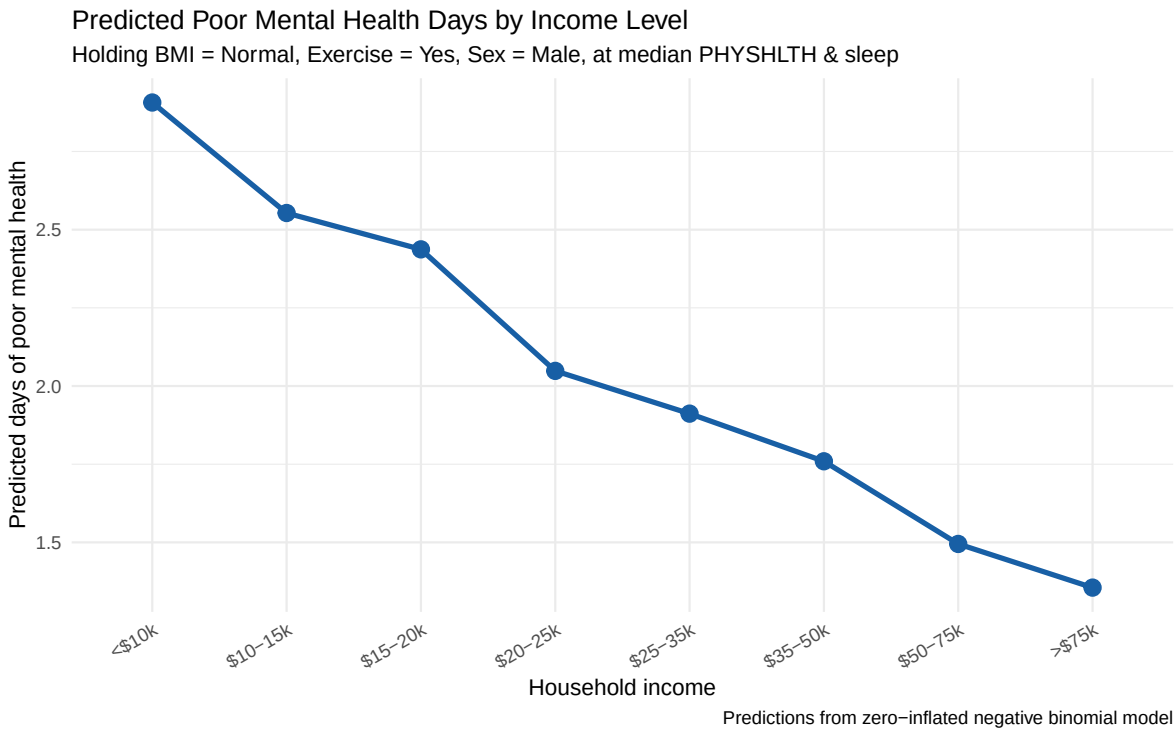
7.2 The Income Gradient

The most striking finding is a near-perfect monotonic income gradient in poor mental health days. Holding all other variables constant, respondents in the lowest income bracket (<$10k/year) reported **2.15× as many** poor mental health days as those earning more than $75k (IRR = 2.145, 95% CI: 1.839–2.501). The effect decreases steadily across income levels, with the $50–75k group being the only income category whose confidence interval overlaps 1 (IRR = 1.103, 95% CI: 0.996–1.221).



7.3 Predicted Days by Income

To put the income effect in interpretable terms, the plot below shows model-predicted poor mental health days across income levels, holding BMI at Normal, exercise at Yes, sex at Male, and physical health and sleep at their sample medians.



7.4 Summary of Key Findings

The table below summarizes all significant IRRs from the count component of the final ZINB model.

Table 3: Incidence Rate Ratios from Zero-Inflated Negative Binomial (Count Component)

Predictor	IRR	95% CI	Significant
Income <$10k (vs. >$75k)	2.145	(1.839, 2.501)	Yes
Income $10-15k (vs. >$75k)	1.884	(1.624, 2.185)	Yes
Income $15-20k (vs. >$75k)	1.798	(1.571, 2.058)	Yes
Income $20-25k (vs. >$75k)	1.511	(1.331, 1.716)	Yes
Income $25-35k (vs. >$75k)	1.411	(1.249, 1.592)	Yes
Income $35-50k (vs. >$75k)	1.298	(1.168, 1.443)	Yes
Income $50-75k (vs. >$75k)	1.103	(0.996, 1.221)	No
Income Underweight (vs. >$75k)	1.115	(0.859, 1.449)	No
Income Overweight (vs. >$75k)	1.058	(0.975, 1.147)	No
Income Obese (vs. >$75k)	1.150	(1.058, 1.25)	Yes
No exercise (vs. active)	1.303	(1.204, 1.409)	Yes
Female (vs. male)	1.133	(1.056, 1.215)	Yes

Predictor	IRR	95% CI	Significant
Physical health days	1.036	(1.032, 1.04)	Yes
Sleep hours	0.938	(0.92, 0.957)	Yes

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8. Discussion

8.1 Income as the Dominant Predictor

The clearest and most consistent finding is the income gradient in poor mental health days. Across all seven income categories below $75k/year, respondents reported significantly more poor mental health days than the highest-income reference group, and the effect size increased steadily as income decreased. The magnitude at the bottom of the income distribution, over twice as many poor mental health days, is substantial and consistent with a large body of prior research linking economic insecurity to mental health burden.

8.2 Behavioral Predictors

Physical inactivity was associated with approximately 30% more poor mental health days (IRR = 1.303), and this effect was statistically significant. This finding is consistent with the existing literature on exercise and mental health, though the cross-sectional design of BRFSS means we cannot determine whether inactivity leads to worse mental health or whether people with worse mental health are less likely to exercise.

Each additional hour of sleep per night was associated with a 6.2% reduction in poor mental health days (IRR = 0.938). While modest in isolation, the effect is meaningful at the population scale and adds to evidence supporting sleep as a behavioral lever for mental wellbeing.

8.3 Physical and Demographic Predictors

Each additional day of poor physical health was associated with a 3.6% increase in poor mental health days (IRR = 1.036). This effect is modest per unit but, given that PHYSHLTH ranges from 0 to 30, the cumulative effect across the full range is meaningful. Female respondents reported 13% more poor mental health days than male respondents (IRR = 1.133), consistent with well-documented sex differences in self-reported mental health.

Of the BMI categories, only obesity was statistically significant (IRR = 1.150), with overweight and underweight categories showing confidence intervals that included 1.

8.4 Limitations

Several limitations should be noted. First, BRFSS is a cross-sectional survey, so none of the associations reported here can be interpreted as causal. Second, all variables, including the response, are self-reported and subject to recall and social desirability bias. Third, the Pearson goodness-of-fit ratio was elevated at 14.68, suggesting some residual misspecification that is common at large sample sizes. Fourth, a natural modeling extension, a mixed-effects negative binomial with state as a random effect, was not feasible due to computational constraints; geographic clustering is therefore unaccounted for in the model. Finally, the slight underprediction at MENTHLTH = 30 suggests a potential boundary effect that the ZINB model does not fully capture.

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9. Conclusion

This analysis modeled the number of self-reported poor mental health days among U.S. adults using a zero-inflated negative binomial GLM fit to BRFSS survey data. Model selection via AIC strongly favored ZINB over both Poisson and plain negative binomial, consistent with the severe overdispersion and high zero rate (69.8%) in the response variable.

The dominant finding is a robust, monotonic income gradient: Americans in the lowest income bracket report more than twice as many poor mental health days as those in the highest bracket, holding health behaviors and demographics constant. Physical inactivity, poor physical health, female sex, obesity, and reduced sleep were each independently associated with more poor mental health days, though income remained the largest single predictor in magnitude.

These results highlight income as a structural determinant of mental health burden at the population level and underscore the importance of behavioral factors, particularly physical activity and sleep, as potentially modifiable contributors.